Chi-Square Tests and Tests for a Population Variance

Chapters 12 and 13

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Objectives

- Explain the chi-square statistic as a measure of how far observed counts sit from the counts a hypothesis predicts, and find its p-value.
- Test whether **k proportions** are equal, and recognize the 2×2 case as last week's two-proportion test.
- Test whether counts follow a stated distribution (**goodness of fit**), including the degrees-of-freedom cost of estimating a parameter.
- Test whether two categorical variables are **independent**, find which cells drive a rejection, and measure how strong the association is.
- Know when expected counts are too small, and how to combine categories.
- Test a single population variance, and build an interval for σ.
- Compare two population variances with the F test.
- Explain why the variance tests need normal data much more than the t-test does, and what to do instead of pretesting variances.

> Weeks 9 and 10 tested claims about **means and proportions**. This week has two new questions.
>
> **Chapter 12: do counts match a hypothesis?** Is the share of "up" days the same every weekday? Do absences follow a Poisson model? Does the brand a customer buys depend on age? Every one of these is answered with one statistic,

$$\chi^2 = \sum_{\text{cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}},$$

and the three tests in the chapter differ only in **where the expected counts come from**. Learn the statistic once, then learn three ways of filling in "expected."

Chapter 13: is the spread what was claimed? A refrigerator can average 37°F and still freeze the lettuce. A checkout process can average 4 minutes and still leave some customers waiting 10. Here the parameter of interest is the variance σ^2, and the test has an important catch that the t-test does not.

Part I: Chi-Square Tests (Chapter 12)

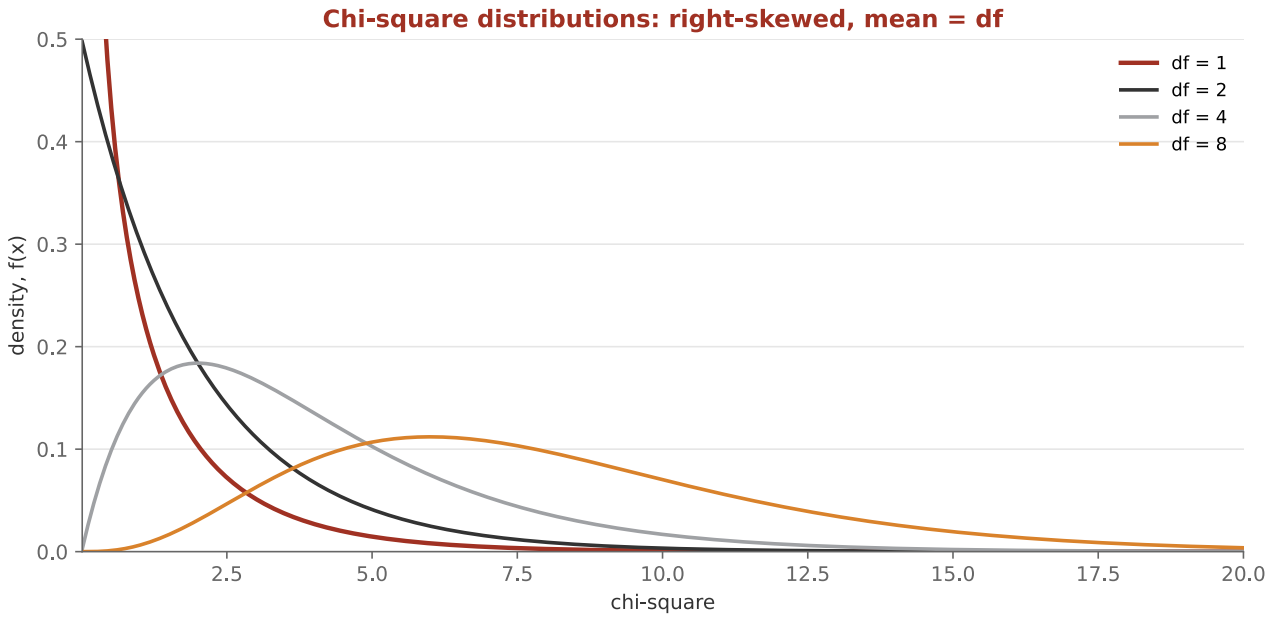
The chi-square statistic

For counts in k cells, the **chi-square statistic** is

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e},$$

where f_o is the **observed** count in a cell and f_e is the **expected** count if H_0 is true. Expected counts need not be whole numbers; they are averages.

If H_0 is true, χ^2 follows (approximately) a **chi-square distribution**, which is right-skewed, never negative, and has mean equal to its degrees of freedom. The approximation needs **every expected count to be at least 5**; if one is not, combine categories.

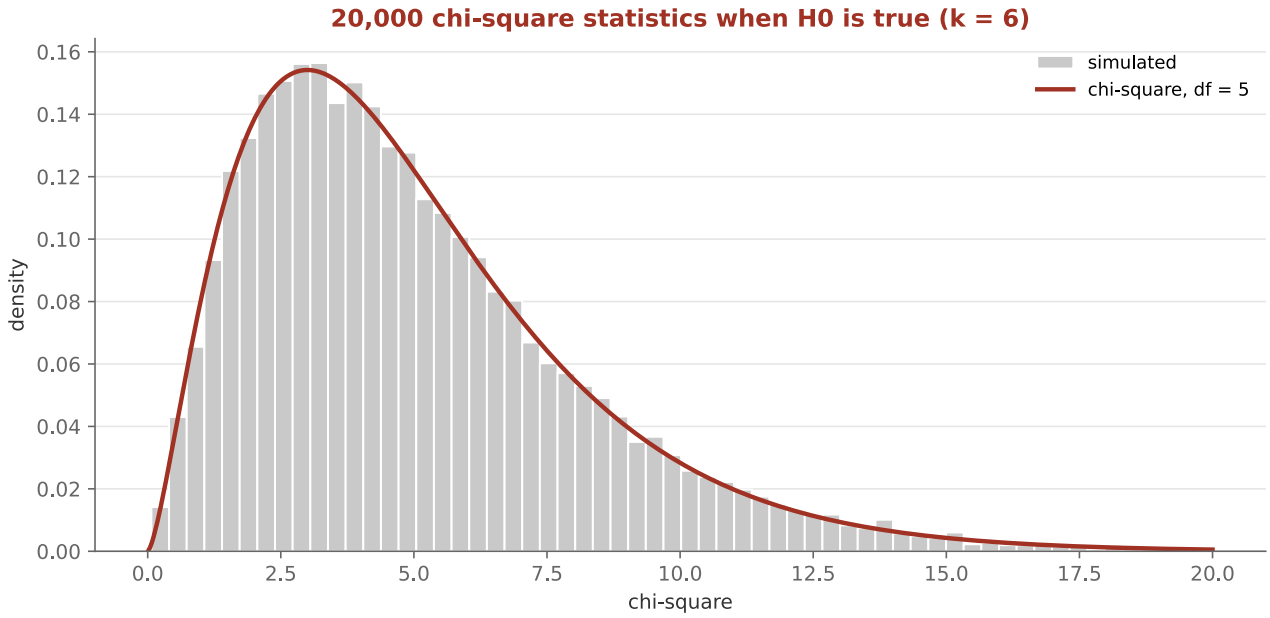


Three things follow from the formula.

- **Every test is upper-tailed.** Squaring makes a cell count too high and a cell count too low both add to χ^2. Only a *large* χ^2 is evidence against H_0, so the p-value is always the right-tail area, Excel's `=CHISQ.DIST.RT(x, df)`.

- **Each cell's contribution is scaled by what was expected.** Being 10 off when 20 were expected matters more than being 10 off when 200 were.
- **A quick sanity check:** since the mean is the df, a χ^2 near its df is entirely ordinary. A value of 5 with 4 df is nothing; a value of 30 with 4 df is a lot.

We can check that the chi-square distribution really describes the statistic. Below, a model with six categories is assumed true, 240 observations are drawn from it 20,000 times, and χ^2 is computed each time.



The simulated statistics average 5.04, against the df of 5, and 5.2% of them exceed the 5% critical value. The formula's distribution is the right one, provided the expected counts are large enough. Here the smallest is 24.

Test 1: are k proportions equal?

To test $H_0 : p_1 = p_2 = \cdots = p_k$, pool all the groups to estimate the common proportion $\bar{p}$, and expect $\bar{p} \times n_j$ successes (and $(1 - \bar{p}) \times n_j$ failures) in group j. The statistic has $\mathrm{df} = k - 1$.

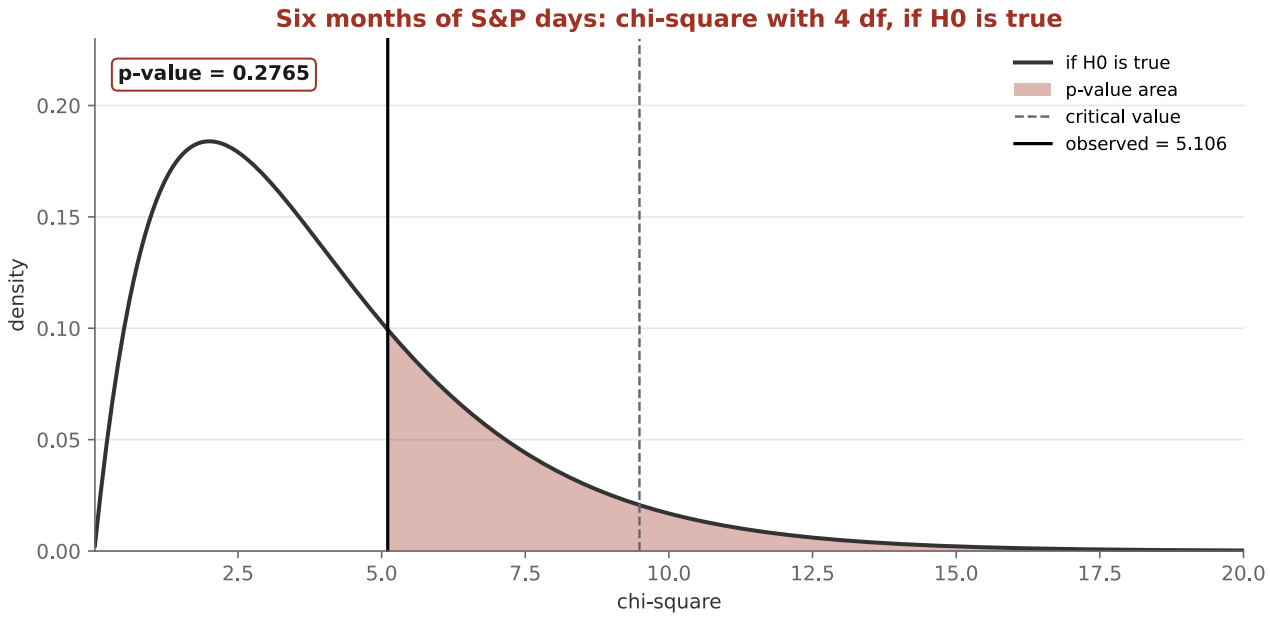
Application: is Monday the best day on Wall Street?

A 2010 *Bloomberg BusinessWeek* article declared that "on Wall Street, Monday is the best day of the week": over six months, the S&P 500 rose on 80% of Mondays, more than on any other weekday. Your textbook tests the article's own data.

	observed f_o	expected f_e	f_o - f_e	(f_o - f_e)^2 / f_e
Monday: success	20	15.600	4.4	1.241
Monday: failure	5	9.400	−4.4	2.060
Tuesday: success	15	16.224	−1.224	0.092
Tuesday: failure	11	9.776	1.224	0.153
Wednesday: success	17	16.224	0.776	0.037

	observed f_o	expected f_e	f_o - f_e	(f_o - f_e)^2 / f_e
Wednesday: failure	9	9.776	−0.776	0.062
Thursday: success	14	15.600	−1.6	0.164
Thursday: failure	11	9.400	1.6	0.272
Friday: success	12	14.352	−2.352	0.385
Friday: failure	11	8.648	2.352	0.640
Total	125	125.000		5.106

	value
chi-square statistic	5.1065
df	4
p-value	0.2765
critical value	9.4877
alpha	0.05
reject H0?	False
Excel p-value	=CHISQ.DIST.RT(5.1065, 4)
Excel critical value	=CHISQ.INV.RT(0.05, 4)
pooled proportion	0.624
smallest expected count	8.648



$\chi^2 = 5.11$ with 4 df, a perfectly ordinary value, and $p = 0.277$. The 80% of Mondays sounds impressive, but with about 25 days per weekday, a gap that size turns up by chance more than a quarter of the time. **The data are not evidence that the share of up days differs by weekday.** (Your textbook gets 5.09; it rounds each expected count before adding.)

The course repository has ten years of daily S&P 500 closes from FRED, August 2016 to August 2026. An up day is a close above the previous day's close.

weekday	up days	trading days	share up
Monday	278	469	0.593
Tuesday	265	518	0.512
Wednesday	286	515	0.555
Thursday	267	506	0.528
Friday	279	505	0.552

	value
chi-square statistic	7.5529
df	4
p-value	0.1094
critical value	9.4877
alpha	0.05
reject H0?	False
Excel p-value	=CHISQ.DIST.RT(7.5529, 4)
Excel critical value	=CHISQ.INV.RT(0.05, 4)
pooled proportion	0.5472
smallest expected count	212.3844

> Over 2,513 trading days, Monday again has the highest share of up days, 59.3%, against 51.2% for the lowest weekday. With twenty times the data, $\chi^2 = 7.55$ and $p = 0.109$. Still no evidence of a weekday effect. A pattern that "everyone knows" and that a reputable magazine printed is, as far as these data can tell, noise. This is the use of statistics most worth taking into a business career.

The 2×2 case is last week's test

With only two groups, the chi-square test of equal proportions **is** the two-proportion z-test of Chapter 10: $\chi^2 = z^2$, with the same p-value. Take last week's email A/B test (216 of 2,400 bought after email B, 168 of 2,400 after email A):

	value
z (Chapter 10)	2.5538
z squared	6.5217
chi-square (this week)	6.5217
p-value, z test	0.0107
p-value, chi-square	0.0107

Same test, same answer. The chi-square version extends to any number of groups; the z version can also be one-tailed ("B is *better*"), which a chi-square test cannot, because squaring throws the direction away.

Test 2: goodness of fit

A **goodness-of-fit test** asks whether observed counts follow a stated probability distribution. H_0 says they do. The expected count in each category is $n \times$ the probability the model gives it, and

$$\mathrm{df} = k - m - 1,$$

where k is the number of categories and m is the number of parameters **estimated from the sample** to compute those probabilities ($m = 0$ if H_0 states the model completely). Each estimated parameter lets the model bend toward the data, and costs one degree of freedom.

Application: a fully stated distribution

A bank's managers believe drive-through demand is spread across the week as 20%, 10%, 10%, 15%, 25%, 20% (Monday to Saturday). One week brings 240 customers. Test at $\alpha = 0.10$.

	observed f_o	expected f_e	f_o - f_e	(f_o - f_e)^2 / f_e
Mon	58	48	10	2.083
Tue	21	24	−3	0.375
Wed	27	24	3	0.375
Thu	20	36	−16	7.111
Fri	71	60	11	2.017
Sat	43	48	−5	0.521
Total	240	240		12.482

	value
chi-square statistic	12.4819
df	5
p-value	0.0287
critical value	9.2364
alpha	0.1
reject H0?	True
Excel p-value	=CHISQ.DIST.RT(12.4819, 5)
Excel critical value	=CHISQ.INV.RT(0.1, 5)
categories k	6
parameters estimated m	0
smallest expected count	24

$\chi^2 = 12.48$ with $6 - 0 - 1 = 5$ df, $p = 0.0287$: **the data are evidence that demand does not follow the stated pattern.** The working table shows where: Thursday contributes more than half of the statistic on its own (20 customers against 36 expected). That is the staffing question to take back to the managers.

Application: a Poisson model with λ estimated

Absences per class meeting in 150 meetings:

students absent	0	1	2	3	4	5	6
class meetings	22	38	39	29	15	5	2

H_0 says absences follow *a* Poisson distribution, without stating its mean. So λ must be estimated from the data, $\bar{x} = 300/150 = 2.0$, and that costs a degree of freedom. The Poisson probability of 6 or more is small enough that its expected count falls below 5, so "5" and "6 or more" are merged into "5 or more."

	observed f_o	expected f_e	f_o - f_e	(f_o - f_e)^2 / f_e
0	22	20.300	1.7	0.142
1	38	40.601	−2.601	0.167
2	39	40.601	−1.601	0.063
3	29	27.067	1.933	0.138
4	15	13.534	1.466	0.159
5 or more	7	7.898	−0.898	0.102
Total	150	150.000		0.771

	value
lambda	2
lambda estimated from data?	True
chi-square statistic	0.771
df	4
p-value	0.9423
critical value	9.4877
alpha	0.05
reject H0?	False
Excel p-value	=CHISQ.DIST.RT(0.7710, 4)
Excel critical value	=CHISQ.INV.RT(0.05, 4)
categories k	6
parameters estimated m	1
smallest expected count	7.898

$\lambda = 2.0$, six categories, one estimated parameter: $\mathrm{df} = 6 - 1 - 1 = 4$. $\chi^2 = 0.77$ and $p = 0.94$. **The data are consistent with a Poisson model.**

Two cautions.

1. **A large p-value does not show that the model is right.** It shows the data cannot tell the model apart from the truth. Using the Poisson to plan an attendance policy may be a sensible business decision, but this test did not prove the data are Poisson. Your textbook goes further than the evidence when it says there is "more evidence supporting the use of the Poisson distribution than there is to not use it."
2. **Excel's `CHISQ.TEST` gets the df wrong here.** `=CHISQ.TEST(observed, expected)` always uses $k - 1$ degrees of freedom. That is right for the drive-through ($m = 0$) and wrong whenever a parameter was estimated: here it would use 5 df instead of 4. Use `=CHISQ.DIST.RT(chi-square, k-m-1)`.

Southwest's late flights. Southwest flies four flights a day between Philadelphia and Orlando. Over 200 days, the number of late flights per day was 0 on 70 days, 1 on 78 days, 2 on 40 days, and 3 or 4 on 12 days. Nationally, 20% of Southwest flights are late. **(a)** Test at $\alpha = 0.10$ whether late flights per day follow a binomial distribution with $n = 4$ and $p = 0.2$. **(b)** How many degrees of freedom, and why? **(c)** The model fails. Look at *where* it fails, and suggest why. (Hint: which binomial assumption does the weather violate?)

Solution

	observed f_o	expected f_e	f_o - f_e	(f_o - f_e)^2 / f_e
0	70	81.92	−11.92	1.73
1	78	81.92	−3.92	0.19
2	40	30.72	9.28	2.80
3 or 4	12	5.44	6.56	7.91
Total	200	200.00		12.64

(a) $\chi^2 = 12.64, p = 0.0055$: reject. The counts do not follow Binomial(4, 0.2). **(b)** Four categories after merging "3" and "4" (the expected count for 4 alone is 0.32); p was stated, so $m = 0$ and $\mathrm{df} = 4 - 0 - 1 = 3$. **(c)** There are too many days with **no** late flights (70 against 82 expected) and far too many with **3 or 4** (12 against 5.4). The binomial assumes the four flights are *independent*. They are not: a thunderstorm in Orlando delays every flight that day, and a clear day delays none. Dependence between trials pushes days toward the extremes, which is exactly the pattern here. Your textbook concludes only that Southwest "needs to develop another model," and its sentence is garbled ("By rejecting the null hypothesis, we have no evidence to conclude that … follows the binomial"). Rejecting H_0 *is* the evidence that the binomial does not fit.

Is it normal? Your textbook also tests normality this way, by cutting the z-scale into intervals (Section 12.2). With 40 observations it can afford only four intervals, leaving **1 degree of freedom**, and such a test will miss almost any departure from normality. Read the section for the method. In practice, look at the data: the tail check from Chapter 6 (what share of values lie beyond ± 2 and ± 3 standard deviations, against the normal's 4.6% and 0.27%), a histogram, or a normal probability plot. Part II shows why this matters.

Test 3: are two categorical variables independent?

A **contingency table** cross-classifies n observations by two categorical variables, with r rows and c columns. The **test of independence** has

$$H_0 : \text{the row and column variables are independent,}$$

and, if H_0 is true, the expected count in a cell is

$$f_e = \frac{(\text{row total}) \times (\text{column total})}{n}, \qquad \text{df} = (r-1)(c-1).$$

Where the formula comes from. Chapter 4 said events A and B are independent when $P(A \text{ and } B) = P(A)\,P(B)$. Estimate each probability from the margins, $P(\text{row}) = \text{row total}/n$ and $P(\text{column}) = \text{column total}/n$, and multiply by n: $n \cdot \frac{\text{row}}{n} \cdot \frac{\text{column}}{n} = \frac{\text{row} \times \text{column}}{n}$. Read it another way: if the column a customer falls in has nothing to do with the row, every row should split across the columns in the same proportions as the whole sample does. The test of independence is Chapter 4's definition, checked with data.

Application: does camera brand depend on age?

A Best Buy store records the brand and the age group of 150 digital camera buyers.

age	Canon	Nikon	Sony	Total
18-34	30	16	8	54
35-51	22	25	19	66
52+	8	9	13	30
Total	60	50	40	150

$$f_e(\text{18-34, Canon}) = \frac{54 \times 60}{150} = 21.6$$

The expected counts for every cell, if brand and age are independent:

age	Canon	Nikon	Sony
18-34	21.6	18.0	14.4
35-51	26.4	22.0	17.6
52+	12.0	10.0	8.0

	value
chi-square statistic	12.1455
df	4
p-value	0.0163
critical value	9.4877
alpha	0.05
reject H0?	True
Excel p-value	=CHISQ.DIST.RT(12.1455, 4)

	value
Excel critical value	=CHISQ.INV.RT(0.05, 4)
rows r	3
columns c	3
n	150
smallest expected count	8
Cramer's V	0.2012

Every expected count is at least 8, so the approximation is fine. $\chi^2 = 12.15$ with $(3-1)(3-1) = 4$ df, $p = 0.0163$: **the data are evidence that brand choice and age are related.**

Where is the dependence? Standardized residuals

A rejection says only that *somewhere* the table departs from independence. To find where, look at each cell's **standardized residual**,

$$\frac{f_o - f_e}{\sqrt{f_e}}.$$

Squared, these are exactly the pieces that add up to χ^2. Read them like z-scores: beyond about ± 2 is a cell that clearly departs from independence, and beyond ± 1.5 is worth a look.

age	Canon	Nikon	Sony
18-34	1.81	−0.47	−1.69
35-51	−0.86	0.64	0.33
52+	−1.15	−0.32	1.77

Three cells carry the result. Buyers aged 18-34 choose Canon more often than independence predicts (residual +1.81) and Sony less often (-1.69); buyers 52 and over choose Sony more often (+1.77). The middle age group looks like the store as a whole. That is the sentence a category manager can use: where to put the Canon display, and who the Sony promotion should target.

How strong is the dependence? Cramér's V

The χ^2 statistic grows with n: double every count and χ^2 doubles, though the pattern is unchanged. So, as in weeks 9 and 10, a small p-value says the association is *real*, not that it is *large*. The effect size for a contingency table is **Cramér's V**:

$$V = \sqrt{\frac{\chi^2}{n\left(\min(r, c) - 1\right)}},$$

which runs from 0 (independence) to 1 (each row predicts the column perfectly). A rough guide for tables with a shorter side of 2 or 3: about 0.1 is small, 0.3 medium, 0.5 large.

$$V = \sqrt{\frac{12.145}{150 \times (3-1)}} = 0.201$$

Now a case where n is large. Your textbook's *Focus on Analytics* box reports a survey by Soliton Technologies of 2,069 people: were they the decision maker the last time their household bought a car?

	Male	Female
Decision maker: yes	207	67
Decision maker: no	811	984

	value
chi-square statistic	87.7024
df	1
critical value	3.8415
alpha	0.05
reject H0?	True
Excel p-value	=CHISQ.DIST.RT(87.7024, 1)
Excel critical value	=CHISQ.INV.RT(0.05, 1)
rows r	2
columns c	2
n	2069
smallest expected count	134.8149
Cramer's V	0.2059

> $\chi^2 = 87.7$ with 1 df, and the p-value is too small for Excel to show (well below one in a billion). Yet $V = 0.21$, about the same as the Best Buy table, which had $p = 0.016$. The two associations are about **equally strong**. The Soliton p-value is tiny because n is 14 times larger. Always report the effect size next to the p-value.

Application: a physician "upcoding" check

Your textbook's box *Using the Chi-Square Test in Health Care* describes Health Care Associates (HCA), which places hospitalists at University Hospital. Patients are assigned to physicians at random, so a physician's patients should have the same mix of admission levels (Level 3 is the most complex, and the most expensive to bill). Insurers watch for **upcoding**: classifying a Level 1 admission as Level 2 or 3 to raise revenue.

	Phys 1	Phys 2	Phys 3	Phys 4	Phys 5	Phys 6
Level 1	3	4	9	9	13	8
Level 2	15	8	21	16	13	19
Level 3	45	27	13	16	19	32

	value
chi-square statistic	32.7164
df	10

	value
p-value	0.0003
critical value	18.307
alpha	0.05
reject H0?	True
Excel p-value	=CHISQ.DIST.RT(32.7164, 10)
Excel critical value	=CHISQ.INV.RT(0.05, 10)
rows r	3
columns c	6
n	290
smallest expected count	6.1862
Cramer's V	0.2375

	Phys 1	Phys 2	Phys 3	Phys 4	Phys 5	Phys 6
Level 1	−2.21	−0.88	0.83	0.98	2.19	−0.44
Level 2	−1.12	−1.24	1.99	0.83	−0.34	0.07
Level 3	2.08	1.45	−2.01	−1.18	−0.94	0.19

$\chi^2 = 32.72$ with $(3-1)(6-1) = 10$ df, $p = 0.0003$. Admission level is not independent of physician. The residuals point at **Physician 1**: 45 Level 3 patients against 33 expected (+2.08), and only 3 Level 1 patients against 10 (-2.21). Physicians 3 and 5 lean the other way. The test does not prove anyone upcoded: the assignment may not be as random as HCA thinks, or Physician 1 may take the night shift when the sickest patients arrive. What it does is tell the compliance office **which chart to pull first**. That is the business use of residuals.

Application: small expected counts, and combining categories

The course's faculty file has the rank and sex of 215 economics professors at nine Big Ten universities in 1999. Is rank independent of sex?

sex rank	Female	Male
Assistant	4	16
Associate	7	33
Professor	8	147

sex rank	Female	Male
Assistant	1.77	18.23
Associate	3.53	36.47
Professor	13.70	141.30

Stop before computing a p-value. The expected counts for female assistant and associate professors are 1.77 and 3.53, below 5. With counts this small, the chi-square distribution is a poor approximation to the statistic's real distribution, and the p-value it gives (0.0090, as it happens) cannot be trusted. Software will compute it anyway, sometimes with a warning, sometimes not.

The fix is to **combine categories** that belong together. Assistant and associate professors are both below full professor, so merge them:

	observed		expected	
sex	Female	Male	Female	Male
rank				
Assistant or Associate	11	49	5.3	54.7
Professor	8	147	13.7	141.3

	value
chi-square statistic	9.3158
df	1
p-value	0.0023
critical value	3.8415
alpha	0.05
reject H0?	True
Excel p-value	=CHISQ.DIST.RT(9.3158, 1)
Excel critical value	=CHISQ.INV.RT(0.05, 1)
rows r	2
columns c	2
n	215
smallest expected count	5.3023
Cramer's V	0.2082

Every expected count is now at least 5.3, and with 1 df, $\chi^2 = 9.32$, $p = 0.0023$. Women are 18.3% of assistant and associate professors but 5.2% of full professors. The data are evidence that rank and sex are related in these departments. Combine on meaning (adjacent ranks, adjacent counts, the top values of a count variable), never by searching for the merge that gives the smallest p-value.

Association is not causation. Every table in this section is observational, so the test of independence says the variables *move together*, not *why*. Your textbook says so correctly. The faculty table shows how much this matters. One explanation is that women were promoted less often. Another is **timing**: if women entered economics faculties in larger numbers only in the 1980s and 1990s, many would not yet have had the years to reach full professor in 1999. The chi-square test cannot tell these apart; data on hiring year and promotion would be needed. The same caution applies to Best Buy (age might stand in for income or for experience with a brand) and to HCA.

Three tests, one statistic.

Test	H_0	Expected count f_e	df
k proportions	$p_1 = \cdots = p_k$	$\bar{p}\, n_j$ and $(1-\bar{p})\, n_j$	$k-1$
Goodness of fit	the counts follow a stated model	$n \times$ model probability	$k-m-1$
Independence	rows and columns are independent	row × column / n	$(r-1)(c-1)$

In all three: $\chi^2 = \sum (f_o - f_e)^2 / f_e$, the p-value is the right tail, and every f_e should be at least 5. The k-proportions test is in fact a test of independence on a $2 \times k$ table (success/failure by group), so the first and third rows always give the same answer.

Part II: Tests for a Population Variance (Chapter 13)

One variance

If the population is **normal**, the statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

follows a chi-square distribution with $n-1$ df when $H_0 : \sigma^2 = \sigma_0^2$ is true. Unlike Chapter 12, the test can be lower-tailed, upper-tailed, or two-tailed, depending on H_1. A claim about a standard deviation is tested as a claim about its square: $\sigma < 2$ minutes is $\sigma^2 < 4$ minutes².

The intuition: s^2/σ_0^2 is the ratio of the variance you saw to the variance claimed, and should be near 1 if the claim is right. Multiplying by $n-1$ puts it on the chi-square scale, whose mean is $n-1$. A statistic far below $n-1$ says the data are less variable than claimed; far above says more.

Application: self-serve checkout (lower tail)

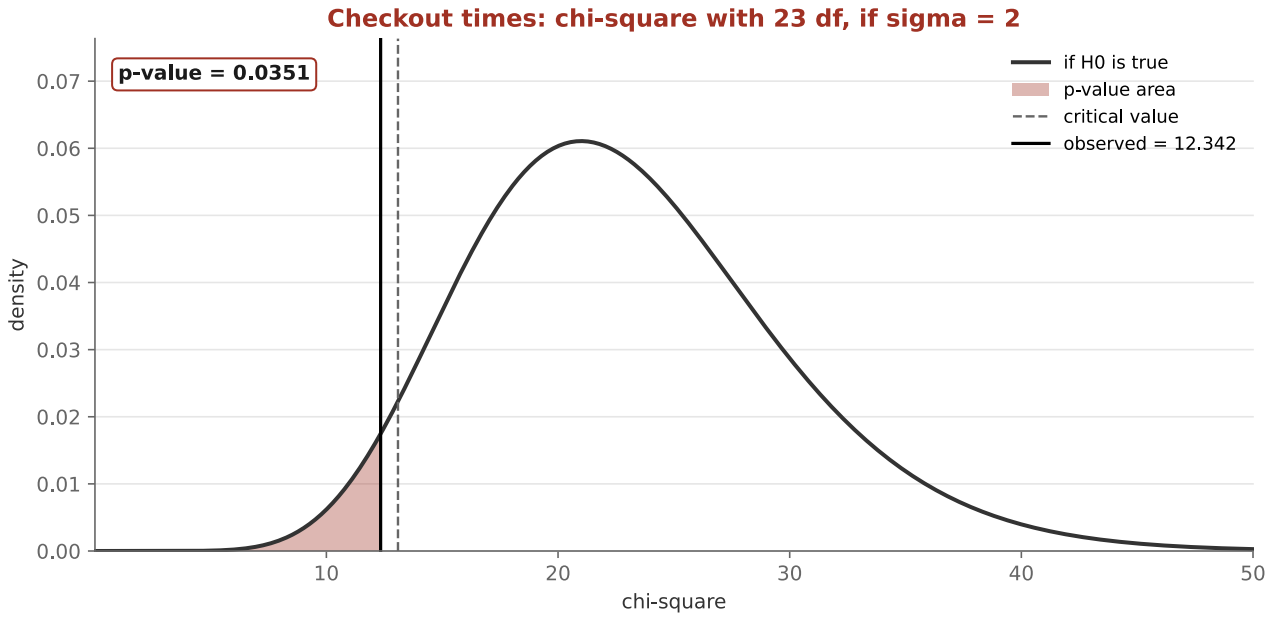
A grocery chain will install self-serve checkout in more stores if the standard deviation of self-serve checkout times is **less than 2 minutes**: a large standard deviation would mean some customers struggle with the machines. A pilot store times 24 self-serve customers. Test at $\alpha = 0.05$:

$$H_0 : \sigma^2 \geq 4 \qquad H_1 : \sigma^2 < 4.$$

	value
H0 variance sigma0^2	4
H0 standard deviation	2
alternative	less
n	24
sample variance s^2	2.1464
sample sd s	1.465
chi-square statistic (n-1)s^2/sigma0^2	12.3416
df	23
critical value(s)	13.0905 (reject below)

	value
p-value	0.0351
alpha	0.05
reject H0?	True
Excel p-value	=CHISQ.DIST(12.3416, 23, TRUE)

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{(24-1)(2.1464)}{4.0} = 12.342$$



$s = 1.465$ minutes, and $\chi^2 = 12.34$, well below the df of 23. The lower-tail p-value, Excel's `=CHISQ.DIST(x, 23, TRUE)`, is 0.0351. **The data are evidence that the standard deviation is below 2 minutes**, and by the chain's rule the systems should go into more stores.

Application: refrigerator temperatures (two tails), and an interval for σ

Amana wants a refrigerator model whose temperature has a standard deviation of **4.0°F**: more variation freezes or spoils food, less means the compressor cycles too often. A test unit records 30 temperatures. Test at $\alpha = 0.05$:

$$H_0 : \sigma^2 = 16 \qquad H_1 : \sigma^2 \neq 16.$$

	value
H0 variance sigma0^2	16
H0 standard deviation	4
alternative	two-sided
n	30
sample variance s^2	19.1506
sample sd s	4.3761

	value
chi-square statistic (n-1)s^2/sigma0^2	34.7104
df	29
critical value(s)	16.0471 and 45.7223
p-value	0.4284
alpha	0.05
reject H0?	False
Excel p-value	=2*MIN(CHISQ.DIST(34.7104,29,TRUE), CHISQ.DIST.RT(34.7104,29))

$\chi^2 = 34.71$ sits between the two critical values, and the two-sided p-value (twice the smaller tail) is 0.428. **The data are not evidence that the standard deviation differs from 4.0°F.**

How close to 4.0°F is the standard deviation, then? A confidence interval answers that. Because the chi-square distribution is skewed, the interval is **not** "estimate ± margin":

For a normal population, a $(1-\alpha)$ confidence interval for σ is

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2,\text{ upper}}}} \le \sigma \le \sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha/2,\text{ lower}}}},$$

where the two χ^2 values cut off $\alpha/2$ in each tail with $n-1$ df. In Excel they are `=CHISQ.INV.RT(alpha/2, n-1)` and `=CHISQ.INV(alpha/2, n-1)`.

	value
s	4.3761
n	30.0000
level	0.9500
chi-square (upper)	45.7223
chi-square (lower)	16.0471
lower limit for sigma	3.4852
upper limit for sigma	5.8829
lower limit for sigma^2	12.1465
upper limit for sigma^2	34.6086

Failing to reject is not showing that $\sigma = 4$. Your textbook concludes that "choosing to find support for the null hypothesis … is the appropriate business decision" and that the design "is working properly." The 95% interval says σ could be anywhere from 3.49°F to 5.88°F. At the top of that range, the refrigerator's temperature varies about 50% more than the target. Thirty readings cannot tell a good design from a mediocre one. The business decision might still be to ship, but the honest summary is "we found no problem, and our test could not have found a moderate one." If the decision matters, collect more data, and decide in advance how far from 4.0°F would be unacceptable.

Canon Business Solutions (CBS) will hire more service staff if the standard deviation of customer wait times for a copier repair **exceeds 30 minutes**. A random sample of 20 service calls has $s = 32.4$ minutes. **(a)** State the hypotheses and test at $\alpha = 0.10$. **(b)** Find and interpret the p-value. **(c)** Does $s > 30$ mean CBS should hire?

Solution

	value
H0 variance sigma0^2	900
H0 standard deviation	30
alternative	greater
n	20
sample variance s^2	1049.76
sample sd s	32.4
chi-square statistic (n-1)s^2/sigma0^2	22.1616
df	19
critical value(s)	27.2036 (reject above)
p-value	0.2763
alpha	0.1
reject H0?	False
Excel p-value	=CHISQ.DIST.RT(22.1616, 19)

(a) $H_0 : \sigma^2 \le 900$, $H_1 : \sigma^2 > 900$ (minutes²). $\chi^2 = 19(32.4^2)/900 = 22.16$, below the critical value 27.2036: do not reject. **(b)** $p = 0.276$, from `=CHISQ.DIST.RT(22.16, 19)`. If σ were exactly 30 minutes, a sample standard deviation at least this large would turn up in about 28% of samples of 20. **(c)** No. A sample standard deviation slightly above 30 is what samples from a $\sigma = 30$ population often produce; the data are not evidence that $\sigma > 30$, so by its own rule CBS does not hire. (This is not evidence that $\sigma \le 30$ either.)

Two variances: the F test

For independent random samples from two **normal** populations,

$$F = \frac{s_1^2}{s_2^2}$$

follows an F distribution with $(n_1 - 1, n_2 - 1)$ df when $H_0 : \sigma_1^2 = \sigma_2^2$ is true. You met this distribution in week 10's ANOVA slice. F near 1 means similar spread.

Which population is "1" is fixed by the question, before looking at the data. For $H_1 : \sigma_1^2 > \sigma_2^2$, the p-value is the right tail, `=F.DIST.RT(F, n1-1, n2-1)`. For $H_1 : \sigma_1^2 \neq \sigma_2^2$, it is twice the smaller tail.

Application: does outsourcing make drive-through times more consistent? (one tail)

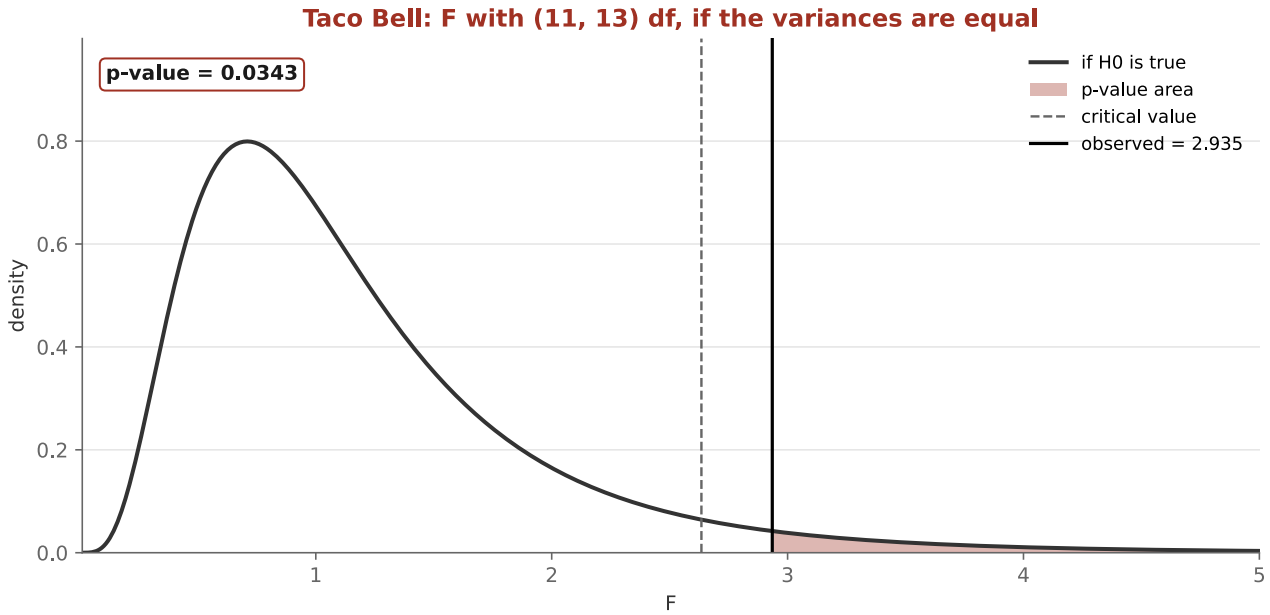
A Taco Bell owner is considering outsourcing drive-through order-taking to a call center. His goal is **more consistent** service times. He times 12 customers under the current on-site system and 14 under a trial of the

outsourced system. The question is whether the outsourced system has the smaller variance, so population 1 is the on-site system:

$$H_0 : \sigma^2_{\text{on-site}} \leq \sigma^2_{\text{outsourced}} \qquad H_1 : \sigma^2_{\text{on-site}} > \sigma^2_{\text{outsourced}}.$$

system	count	mean	std
on-site	12	141.8	46.7
outsourced	14	155.4	27.3

	value
population 1	on-site
population 2	outsourced
alternative	greater
s1^2	2184.9318
s2^2	744.4176
F = s1^2/s2^2	2.9351
df numerator	11
df denominator	13
critical value(s)	2.6347 (reject above)
p-value	0.0343
alpha	0.05
reject H0?	True
Excel upper-tail area	=F.DIST.RT(2.9351, 11, 13)



$F = 2184.9 / 744.4 = 2.935$, $p = 0.0343$. **The data are evidence that the outsourced system gives more consistent service times.** The outsourced mean is, if anything, a little *higher* (155 against 142 seconds). The case for outsourcing is entirely about spread, which a t-test of the means would never see.

Application: hospital stays (two tails)

Blue Cross Blue Shield (BCBS) compares the variability of the length of stay for one procedure at two hospitals, with no prior view of which is more variable. From 20 patients at each: Christiana $s = 0.92$ days, Bayhealth $s = 0.71$ days. Test at $\alpha = 0.05$.

	value
population 1	Christiana
population 2	Bayhealth
alternative	two-sided
s1^2	0.8464
s2^2	0.5041
F = s1^2/s2^2	1.679
df numerator	19
df denominator	19
critical value(s)	0.3958 and 2.5265
p-value	0.2676
alpha	0.05
reject H0?	False
Excel upper-tail area	=F.DIST.RT(1.6790, 19, 19)

$F = 0.92^2/0.71^2 = 1.679$, and the two-sided $p = 0.268$, twice `=F.DIST.RT(1.679, 19, 19)`. **The data are not evidence that the hospitals differ in variability.** Note the units: the test is about variances (days^2), so square the standard deviations first. Your textbook warns about this, and it is the most common error in this section.

Your textbook's rule, "always put the larger sample variance in the numerator and do a right-tail test," is safe only for two-tailed tests. For a two-tailed test, larger-on-top with $\alpha/2$ in the right tail gives exactly the p-value above. For a **one-tailed** test, it lets the data choose the direction, the tail-picking problem from week 9. In the Taco Bell case the question and the data happen to agree. Had the outsourced sample come out *more* variable, the rule would have flipped the populations and tested a question the owner never asked. Set the direction from the question. If the data go the other way, $F < 1$ and you simply do not reject.

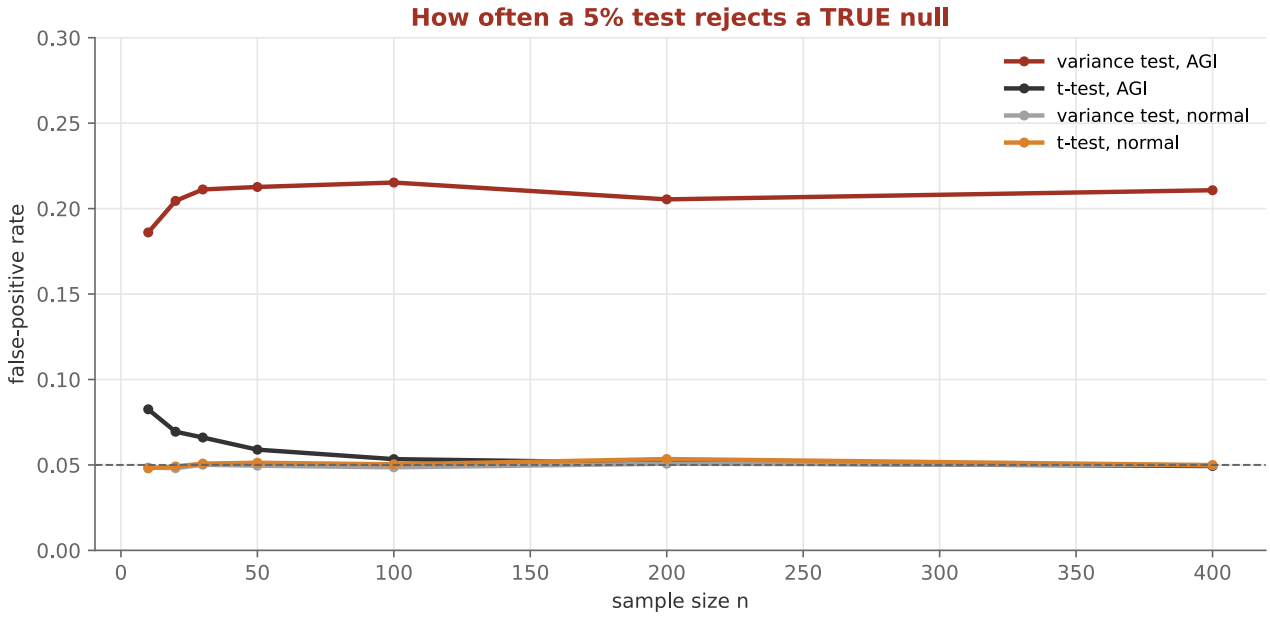
The catch: these tests need normal data, and more data does not help

Every test in weeks 9 and 10 leaned on the Central Limit Theorem: whatever the population's shape, $\bar{x}$ is close to normal once n is large enough. **Nothing like that rescues s^2.** The distribution of $(n-1)s^2/\sigma^2$ is chi-square only for a normal population, and for a skewed or heavy-tailed population it is wrong at every sample size: its spread depends on how heavy the population's tails are.

To see how much this matters, take the course's Nevada income file, 100,000 simulated tax returns (AGI in tens of thousands of dollars), strongly right-skewed. Treat it as the population. Draw 20,000 samples of size n, test the

true variance at $\alpha = 0.05$ two-sided, and count how often the test rejects. Every rejection is a false positive, so a valid test should reject 5% of the time. For comparison, do the same with a t-test of the true mean, and with a normal population.

n	variance test, AGI	t-test, AGI	variance test, normal	t-test, normal
10	0.186	0.083	0.048	0.048
20	0.204	0.069	0.048	0.049
30	0.211	0.066	0.050	0.051
50	0.213	0.059	0.050	0.051
100	0.215	0.053	0.049	0.050
200	0.205	0.051	0.051	0.053
400	0.211	0.049	0.050	0.050



With normal data, both tests reject about 5% of the time, as advertised. With the skewed income data, the t-test starts too high (8.3% at $n = 10$) and settles to about 5% as n grows, the Central Limit Theorem at work. **The variance test does not settle.** It rejects a true null about 21% of the time at $n = 30$ and still 21% at $n = 400$: four times the advertised rate, and a larger sample buys nothing. The F test, built from two sample variances, has the same weakness.

What to do about it.

1. **Use the variance tests only for data you have reason to think are normal:** measurements from a stable process (fill weights, temperatures, machined parts), not incomes, sales, or waiting times with a long right tail. Check first: a histogram, and the Chapter 6 tail check.
2. **Do not use the F test to choose between the pooled and Welch t-tests.** Your textbook suggests it (the rat Your Turn). The F test is the *less* reliable of the two procedures, and choosing a test based on a pretest

changes the error rates of the test you end up running. Use Welch, as in week 10. It is valid whether or not the variances are equal.

3. When a variance question matters on non-normal data, more robust methods exist (Levene's test in most software, or a bootstrap interval). They are beyond our textbook; ask Claude to run one and explain it.

Which Test Is It?

question	data	statistic / expected count	df	tail
Are k proportions equal?	counts of successes in k groups	pooled p-bar times n_j	k - 1	right tail
Do counts follow a stated model?	counts in k categories	n times model probability	k - m - 1	right tail
Are two categorical variables independent?	an r x c table of counts	row x column / n	(r - 1)(c - 1)	right tail
Is a variance equal to a claimed value?	one sample, normal population	(n - 1)s^2 / sigma0^2	n - 1	from H1
Are two variances equal?	two independent samples, normal populations	F = s1^2 / s2^2	(n1 - 1, n2 - 1)	from H1

Name the test, state H_0, and give the degrees of freedom.

(a) A casino records which of four table games 400 guests chose, and whether each guest was a loyalty member or not.

(b) A call center claims 50% of its calls arrive in the morning, 30% in the afternoon, and 20% in the evening. A day's 600 calls are classified.

(c) A bottler's filling machine is specified to have a standard deviation of no more than 0.15 ounces. Thirty bottles are weighed.

(d) Three Las Vegas hotels each survey guests on whether they would return: 180 of 240, 150 of 220, and 200 of 250 say yes.

(e) The number of customer complaints per day over 90 days is tabulated to decide whether a Poisson model can be used for staffing.

i Solution

(a) Independence on a 2×4 table: H_0: game choice is independent of membership; $\mathrm{df} = (2-1)(4-1) = 3$. It cannot show that membership *causes* a game choice. **(b)** Goodness of fit with a stated model: H_0: the shares are 0.5, 0.3, 0.2; $\mathrm{df} = 3 - 0 - 1 = 2$. **(c)** One-variance test, upper tail: $H_0 : \sigma^2 \le 0.0225$ against $H_1 : \sigma^2 > 0.0225$, with 29 df. Fill weights from a stable process are a reasonable place to assume normality, but check a histogram. **(d)** k proportions: $H_0 : p_1 = p_2 = p_3$, $\mathrm{df} = 2$. **(e)** Goodness of fit with λ estimated by $\bar{x}$, so $m = 1$ and $\mathrm{df} = k - 2$ after combining the top categories until every expected count is at least 5. Remember that a large p-value means "no evidence against Poisson," not "Poisson confirmed."

Doing It in Excel

task	Excel	note
chi-square p-value (right tail)	=CHISQ.DIST.RT(x, df)	every Chapter 12 test; upper-tail variance test
chi-square critical value	=CHISQ.INV.RT(alpha, df)	upper critical value
chi-square left-tail area	=CHISQ.DIST(x, df, TRUE)	lower-tail variance test
p-value from observed and expected ranges	=CHISQ.TEST(obs, exp)	uses df = k - 1 (one column) or (r-1)(c-1): WRONG if parameters were estimated
sample variance	=VAR.S(range)	s^2, divisor n - 1
F p-value (right tail)	=F.DIST.RT(F, df1, df2)	double it for a two-tailed test
F critical value	=F.INV.RT(alpha, df1, df2)	use alpha/2 for a two-tailed test
F test from raw data	Data Analysis > F-Test Two-Sample for Variances	reports ONE-tailed results only
F test p-value from raw data	=F.TEST(range1, range2)	TWO-tailed p-value

Two Excel traps. `CHISQ.TEST` returns a p-value directly from the observed and expected ranges, but it always uses $k-1$ df for a single column (or $(r-1)(c-1)$ for a table), so it is wrong for a goodness-of-fit test in which a parameter was estimated. And the **Data Analysis → F-Test Two-Sample for Variances** tool reports one-tailed results only, for the populations in the order you entered them. For a two-tailed test, double its p-value, or use `=F.TEST(range1, range2)`, which is already two-tailed.

Reading the Textbook: Corrections

1. **"A smaller chi-square test statistic … increases the likelihood that there is support for the null hypothesis"** (Section 12.1). A small χ^2 means the data fail to contradict H_0. It is not support for H_0.
2. **The drive-through example sets $\alpha = 0.10$**, then compares the p-value with 0.05 in its last step. The conclusion is the same either way, but use the α you set.
3. **Southwest's late flights.** "By rejecting the null hypothesis, we have no evidence to conclude that the number of late flights … follows the binomial distribution" is garbled: rejecting H_0 *is* the evidence that the binomial does not fit. And the book never says why it fails (weather makes a day's flights dependent; see the exercise above).
4. **"More evidence supporting the use of the Poisson distribution than there is to not use it"** ($p = 0.94$), and similarly for the normal fit to SAT scores. A large p-value is not evidence for the model. The normal fit used 1 df on 40 scores and would miss almost any departure.
5. **`CHISQ.TEST` is presented as equivalent to `CHISQ.DIST.RT`.** It is only when no parameter was estimated ($m = 0$).
6. **"Always assign the larger sample variance to Population 1"** (Section 13.2). Fine for a two-tailed test; for a one-tailed test, the direction comes from the question (above).
7. **Using the F test to decide whether to pool** (the rat Your Turn). Don't pretest; use Welch.
8. **The waiting-line *Focus on Analytics* box** gives a variance "of 7.2 minutes" and says times "would vary by less than 7.2 minutes." A variance is in minutes^2; the standard deviation is $\sqrt{7.2} \approx 2.7$ minutes.

9. **Typos.** `CHI.DIST.RT` should be `CHISQ.DIST.RT`. The checkout example's hand-computed variance is misprinted; Excel's `VAR.S` value, 2.1464, is correct.
10. **The refrigerator: "choosing to find support for the null hypothesis."** This is accepting H_0. The interval for σ above shows how little 30 readings pin down.
11. **The book's normality warning is right and understated.** It calls the normal requirement "fairly rigid." The AGI simulation shows a nominal 5% test rejecting a true null about 21% of the time at every sample size.

Laboratory 12–13

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 12–13 pack — one file. Unzip it and you will have a folder holding 11 files. Drag all 11 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

> I have uploaded 11 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.
>
> Then run `12_13_chi_square_and_variance_tests.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 12-13 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.
>
> Then run `12_13_chi_square_and_variance_tests.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

PROMPT

A hotel group surveyed 390 guests about the booking channel they used (direct website, online travel agency, phone, travel agent) and their trip purpose (business, leisure). The counts are: business = 92, 58, 24, 4; leisure = 88, 102, 16, 6 (in the channel order above). By running code, (1) test at alpha = 0.05 whether channel and trip purpose are independent, and check every expected count before trusting the p-value; if any is below 5, combine categories sensibly and test again; (2) show me the standardized residuals and tell me in plain English which cells drive the result; (3) report Cramer's V and say whether the association is weak, moderate, or strong; and (4) tell me whether these data show that trip purpose causes the choice of channel.

Check four things in the reply: the expected count for business travelers booking through a travel agent is below 5, so the reply should combine "phone" and "travel agent" (or explain another sensible merge) before concluding; the residual discussion names specific cells rather than repeating "reject"; Cramér's V is reported with a verbal size; and the answer to the last question says the survey is observational, so it shows association only.

Working in Excel

PROMPT

Build me an Excel workbook that shows why the variance test needs normal data. Sheet 1: in A2:A31, 30 values from a normal population with standard deviation 10, =NORM.INV(RAND(), 50, 10). In B2:B31, 30 values from a skewed population with the same standard deviation 10, =-10*LN(RAND()) (an exponential with mean 10 and standard deviation 10). At the top, for each column, compute the sample variance =VAR.S, the statistic =29*VAR.S(…)/100, and the two-sided p-value =2*MIN(CHISQ.DIST(x, 29, TRUE), CHISQ.DIST.RT(x, 29)), and a cell that says "reject" when the p-value is below 0.05. (Use formulas rather than the Data Analysis tool, whose output does not recalculate.) Give me the file to download.

Press F9 forty times and count how often each column says "reject." Both columns test a **true** null ($\sigma = 10$). Which column rejects close to 5% of the time, and which rejects far more often? In one paragraph, explain what this means for a manager who wants to use the variance test on waiting times or sales.